

Introduction to Artificial Neural Networks

Understanding the "Brain" of AI

CSC212 Artificial Intelligence
Faculty of Emerging Sciences & Technology

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Have you ever wondered how your phone recognizes your face, or how YouTube knows which video you'll like next?

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The answer often involves a Neural Network.

Inspired by the Human Brain

The Human Brain

Our brain is a massive network of cells called neurons.

- Millions of neurons
- Connected by synapses
- Learns from experience

The Neural Network

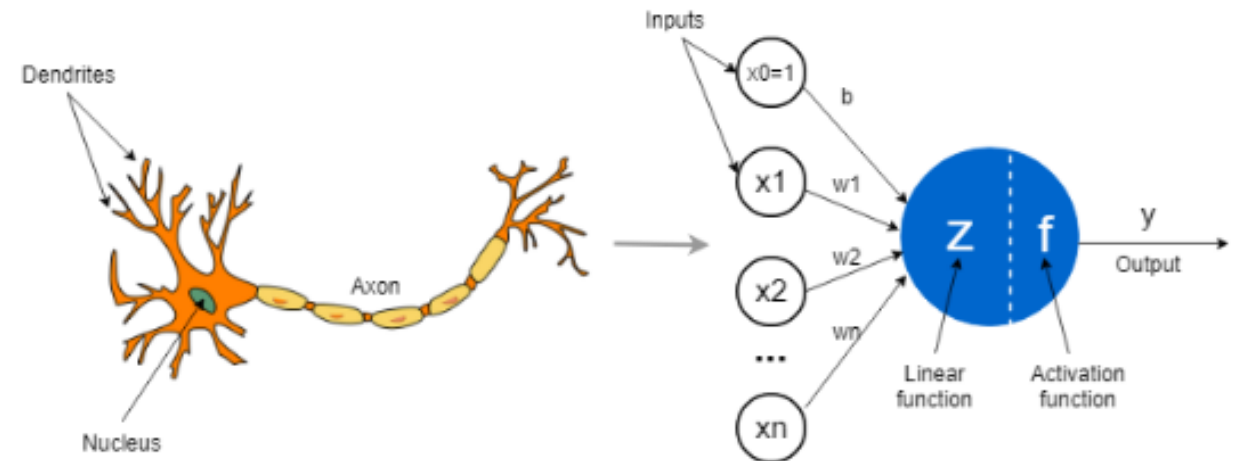
A computer system modeled on the human brain.

- Artificial 'neurons' (nodes)
- Connected by 'weights'
- 'Learns' from data

The Basic Building Block

The simplest unit is called a 'perceptron' or 'neuron'. It takes in information, processes it, and passes on a signal.

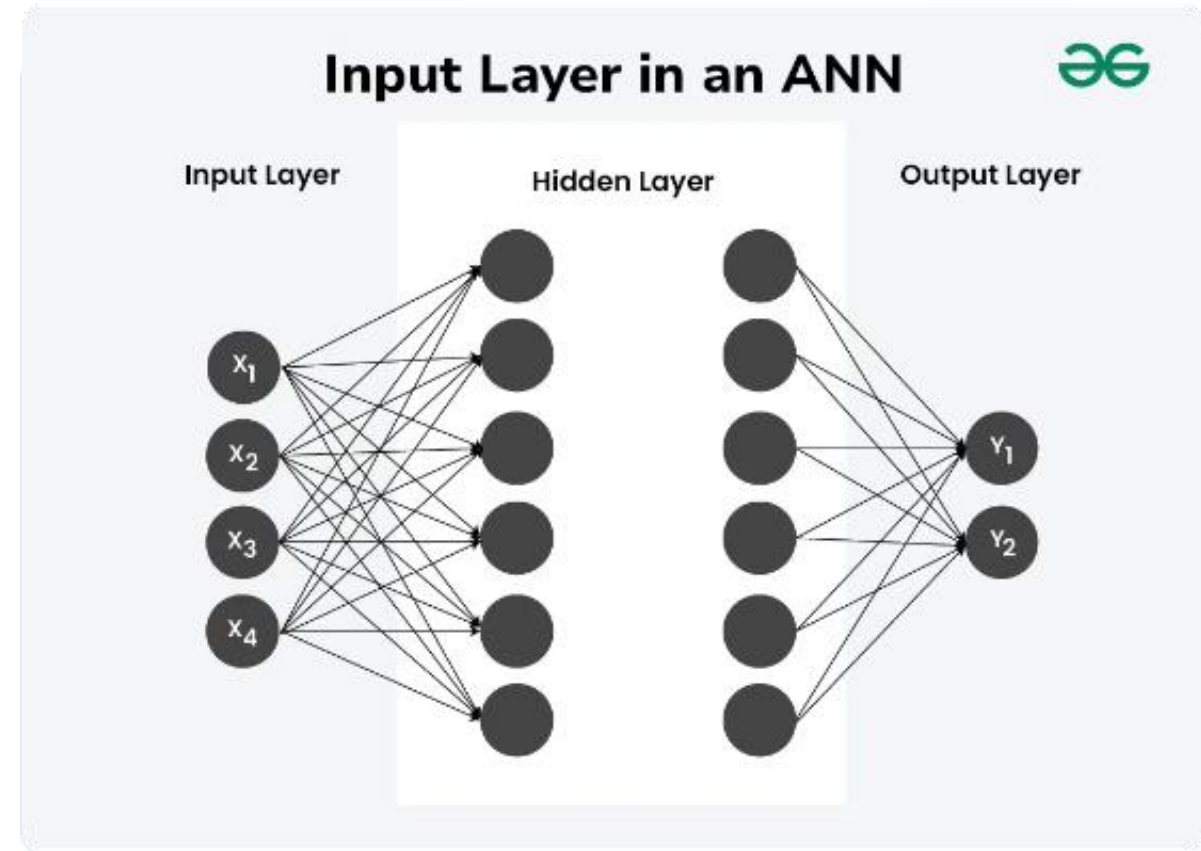
- **Inputs:** Information it receives.
- **Weights:** The 'importance' of each input.
- **Activation:** Decides whether to send a signal.
- **Output:** The final signal it sends.



How They Connect: Layers

Neurons are organized into layers to form a complete network.

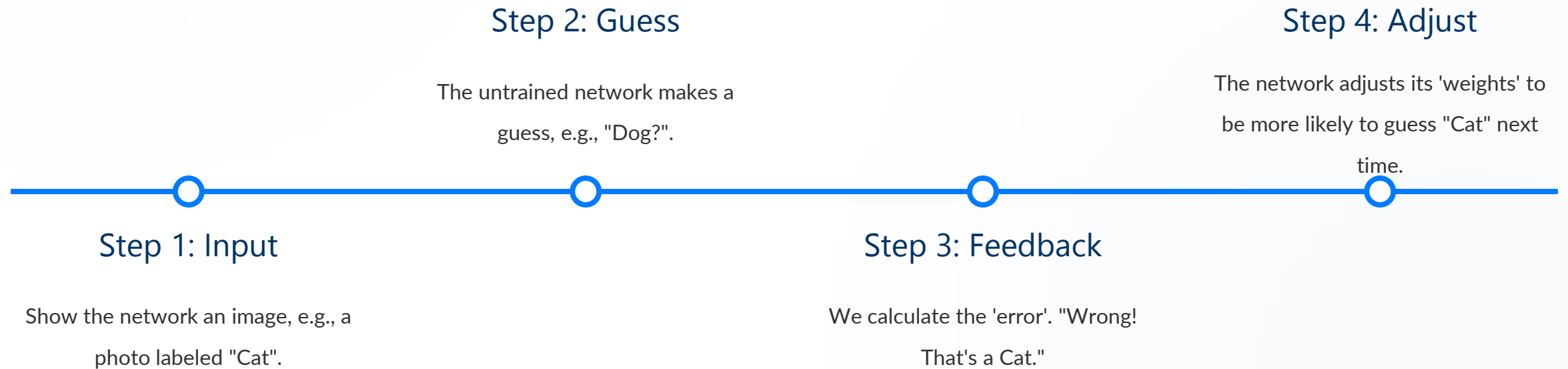
- **Input Layer:** Receives the raw data (like pixels in an image).
- **Hidden Layer(s):** The 'thinking' part. This is where the magic happens!
- **Output Layer:** Gives the final answer (e.g., "Cat" or "Dog").



How Do They "Learn"?

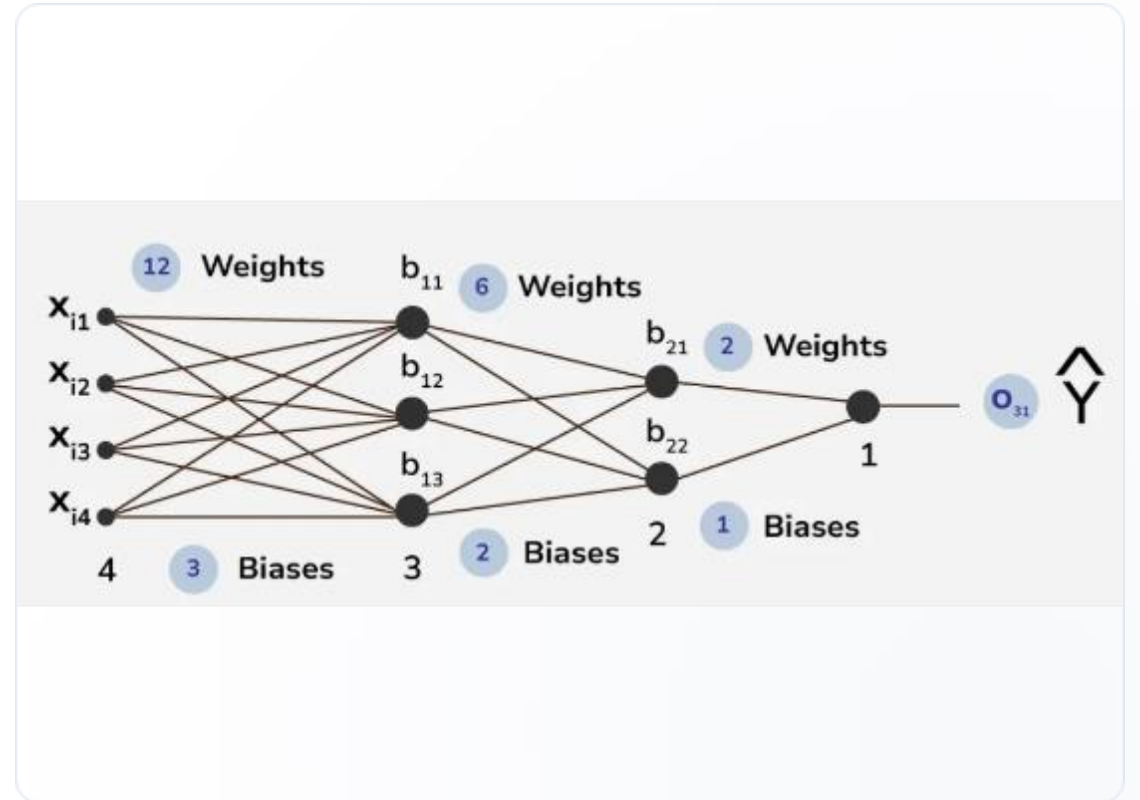
It's all about learning from mistakes.

The Learning Process (Training)



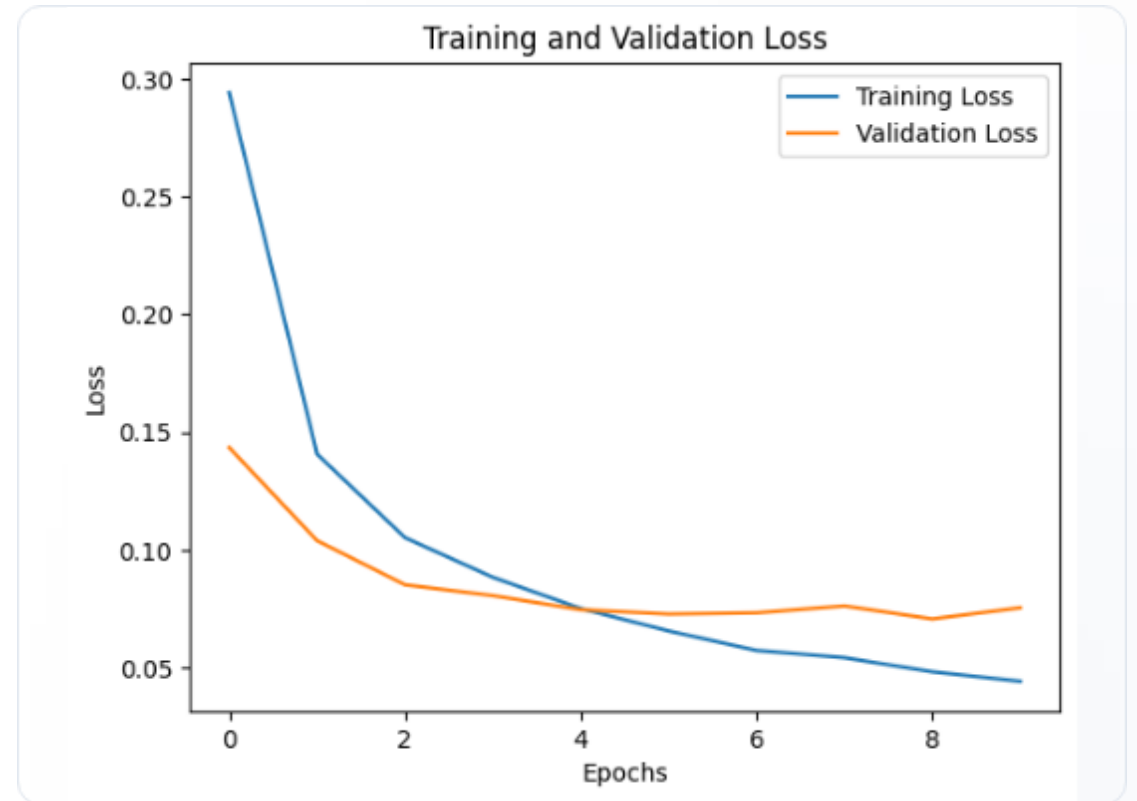
Training: Step 1 & 2 (Forward Pass)

- **Step 1: Initialization:** The network starts with random 'weights'. It knows nothing.
- **Step 2: Forward Propagation:** We feed data (e.g., an image) into the input layer.
- The data "flows" forward, layer by layer, as each neuron performs a calculation.
- The output layer makes its first prediction or "guess".



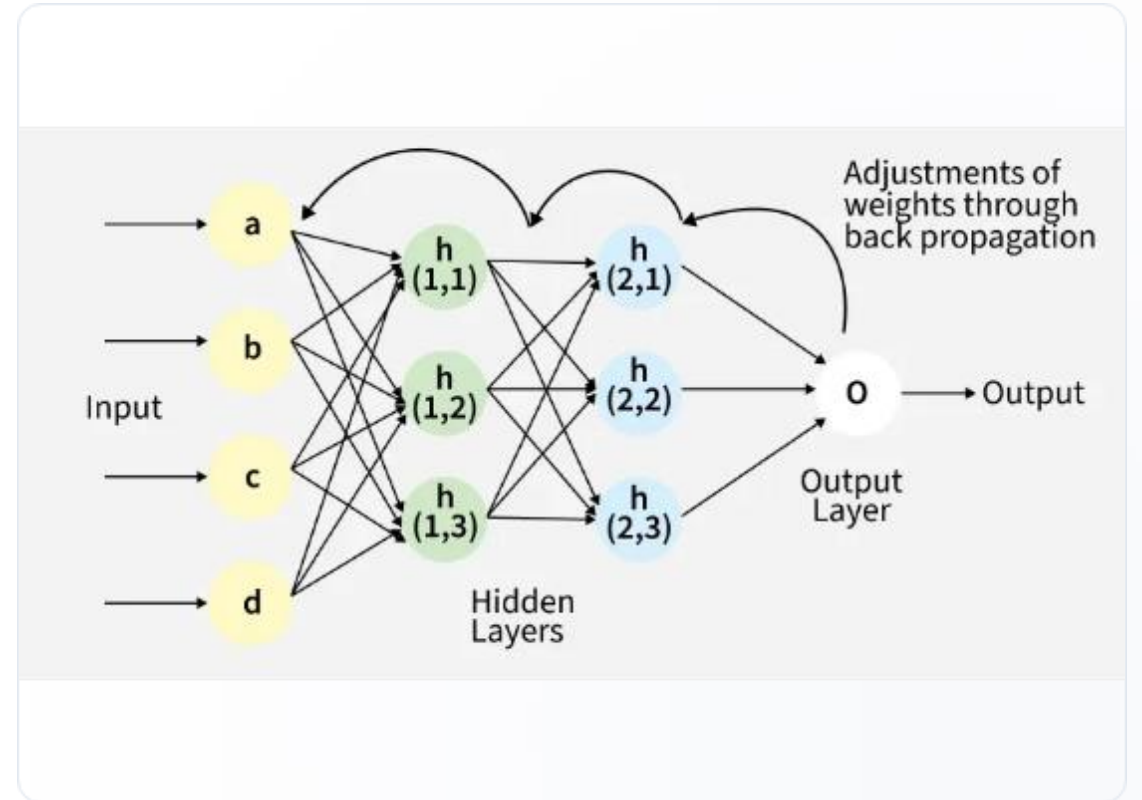
Training: Step 3 (Calculate Error)

- The network compares its guess (e.g., "Dog") with the *actual* answer (e.g., "Cat").
- A "**Loss Function**" measures *how wrong* the guess was.
- This gives us a number called 'loss' or 'error'.
- **The Goal:** Get this error number as close to zero as possible.



Training: Step 4 (Backpropagation)

- This is the "learning" part!
- The network sends the error message *backwards* from the output layer.
- It calculates how much each individual 'weight' (connection) contributed to the error.
- It then adjusts all the weights *just a little bit* to reduce the error.



Training: Step 5 (Repeat!)



Iterate: We repeat this process (Forward Pass -> Calculate Error -> Backpropagation) thousands or millions of times.



Epoch: One full pass through the *entire* training dataset is called one 'epoch'.



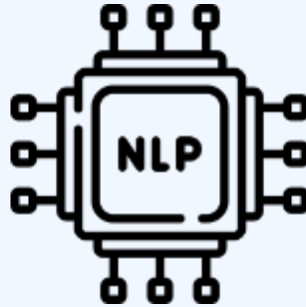
Result: With each iteration, the network's weights get better, the error gets smaller, and the predictions become more accurate.

Applications of ANN



Image Recognition

Tagging photos, unlocking your phone, and powering self-driving cars.



Language

Google Translate, Siri/Alexa, spam filters, and chatbots.



Recommendations

Netflix, YouTube, and "You might also like..." on Amazon.

Examples



CNNs

Convolutional Neural Networks.
These are the 'kings' of image and video recognition.



RNNs

Recurrent Neural Networks. They are great for sequences, like text and speech.



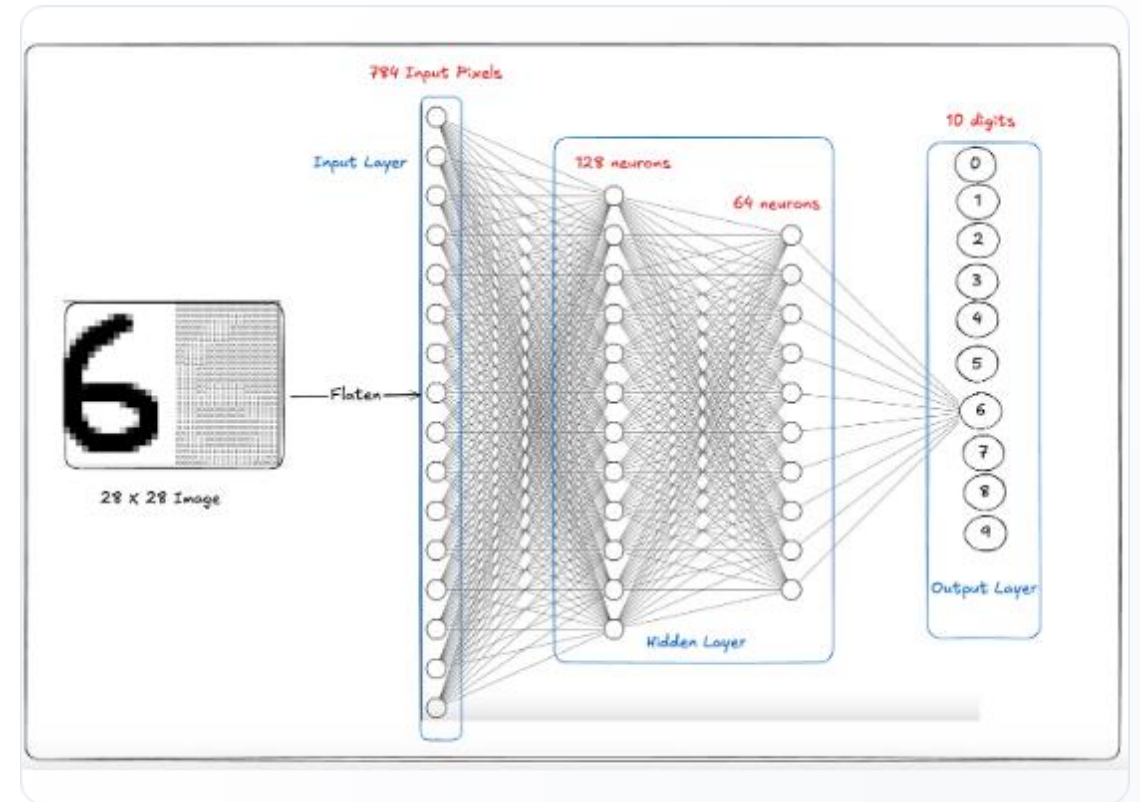
GANs

Generative Adversarial Networks.
They can **create** new, realistic images, music, and more!

Example: Reading Handwriting

How does a network learn to read a handwritten '7'?

- **Input:** The image of a '7' is broken into a grid of pixels.
- **Hidden Layers:** Layers find simple edges, then combine them into shapes (like a horizontal line and a diagonal
- **Output:** The final layer has 10 neurons (0-9). The '7' neuron "lights up" the brightest!



Conclusion



Neural Networks are **inspired by the human brain**.



They are made of 'neurons' connected in **layers**.



They **learn from data** by adjusting 'weights' (connections).



They are masters of **pattern recognition** and power many of the AI tools you use daily.